

TribeLink — Complete App & Feature Guide

"Find Your Tribe" — AI-Powered Interest-Based Community Platform

Generated on: July 30, 2026 | Platform: iOS & Android (Expo / React Native + Supabase)

1. Executive Summary & Core Philosophy

TribeLink is a modern, location-aware, interest-based social platform built for iOS and Android. It operates on the "Online First, Offline Later" philosophy: empowering individuals to discover nearby people who share their exact passions, connect in a safe digital space, build trust through gamified metrics, and meet in real life.

2. Authentication & Identity Management

- **Multi-Method Sign-In:** Supports Email/Password, Google Sign-In, Apple Sign-In, and Phone OTP verification.
- **Biometric Security:** Integrated device biometric authentication (FaceID & Fingerprint).
- **Password Recovery:** Complete forgot-password flow with OTP verification and secure reset screens.
- **Bulletproof User Triggers:** Automatic username generator with fallback safety handling in PostgreSQL.

3. 6-Step Guided Onboarding

- **Step 1: Personal Info:** Setup of full name, avatar, and custom username.
- **Step 2: Profile Photo:** Camera integration and photo picker for avatar customization.
- **Step 3: Interest Matrix:** Selection from 80+ categorized interests with skill level ratings (Beginner, Intermediate, Expert).
- **Step 4: Goals & Skills:** Define personal aspirations, target dates, and skills to share or learn.
- **Step 5: Location Sharing:** PostGIS geolocation setup with private exact coordinates and public fuzzy location.
- **Step 6: Privacy & Preferences:** Configure direct message permissions (Everyone, Connections, None), age visibility, and radius.

4. AI-Powered Discovery & Proximity Matching Engine

- **AI Match Score Algorithm:** Calculates match percentages (0-100%) evaluating distance, shared interests, skills, goals, age, and availability.
- **PostGIS Geolocation:** Proximity search using ST_DWithin and spatial indexing within user-defined kilometer radius.
- **Discovery Modes:** Grid View and List View showing match percentages, mutual interest counts, and distance indicators.
- **Filter Controls:** Customizable distance slider, age range, interest tags, languages, and availability filters.

5. Interest Communities ("Tribes")

- **Community Types:** Public (open), Private (approval required), and Invite-Only (secret/exclusive).
- **Community Hub:** Dedicated feed, member directory, rules tab, pinned posts, and moderator action tools.
- **Role Hierarchy:** Role-based access control supporting Owner, Admin, Moderator, and Member roles.
- **Creation Tools:** Modal wizard to create communities with banners, avatars, categories, tags, and custom rules.

6. Events & Local Meetups

- **Event Formats:** Supports Offline (in-person), Online (virtual), and Hybrid events.
- **Interactive Map:** React Native Maps / Google Maps integration displaying nearby upcoming events.
- **RSVP & Capacity:** Attendance tracking (Going, Maybe, Not Going) with capacity caps and automatic waitlists.
- **QR Code Check-In:** Generates unique QR codes for attendance verification at offline events.

7. Activity Feed & Multi-Media Posts

- **Feed Views:** Filter feed by All, Following, Communities, or Pinned posts.
- **Multi-Media Post Types:** Supports Text, Image/Video galleries, Interactive Polls, Articles, Questions, and Code Snippets.
- **Interactions:** Likes/reactions, nested multi-threaded comments, bookmarks, shares, mentions, and hashtags.

8. Real-Time Direct & Group Messaging

- **Realtime Delivery:** Powered by Supabase Realtime Channels for instant message syncing.
- **Rich Messaging:** Text, Images, Files, and Voice Notes with audio recording and waveform playback.
- **Chat Features:** Read receipts, message reactions, threaded replies, and unread counters.

9. Gamification, XP & Achievements

- **XP & Leveling System:** Levels 1 to 10 unlocked by participating, creating content, and attending events.
- **Collectible Badges:** 10 collectible badges with rarity tiers (Common, Rare, Epic, Legendary).
- **Daily Streaks:** Tracks consecutive active login days to incentivize continuous engagement.

10. Trust Score & Security Engine

- **Trust Score (0-100):** Dynamic rating calculated from profile completeness, verification, and behavior.
- **Location Privacy:** Dual-location architecture: exact location kept strictly private, fuzzy location exposed publicly.
- **Moderation System:** In-app user/post reporting plus Supabase Edge Function automated text and image moderation.

11. Universal Search & Notifications

- **Global Search:** Indexes profiles, posts, communities, and events using PostgreSQL tsvector full-text search.
- **Notification Center:** Real-time push notifications via Expo Push API for chats, mentions, RSVPs, and AI matches.

12. Technology Stack Overview

- **Frontend Mobile:** React Native, Expo (SDK 52+), TypeScript, Expo Router
- **Styling & Theme:** NativeWind (Tailwind CSS), Expo Linear Gradient, Expo Blur
- **Backend & DB:** Supabase (PostgreSQL, Realtime, Auth, Storage, PostGIS, pgvector)
- **State & Forms:** Redux Toolkit, React Query, React Hook Form + Zod

